

ENG/TE 566 Computer Program

Bookkeeping Program for Mr. Truck Inc.

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Overview

The program is a web-based bookkeeping system written in Python (Flask and SQLite). Every business activity is recorded as a balanced double-entry journal entry, and the income statement, balance sheet and cash flow statement are recalculated from the journal after each activity. All invoices, purchase orders, payroll events and paid bills are stored for later review, and every statement can be printed.

Sales and payables are on **net-30 terms**: a sale is recorded as *Accounts Receivable* and an inventory purchase as *Accounts Payable*. Both are settled into cash at month-end close. Payroll is paid from cash immediately.

Initial conditions. Initial investment of \$200,000 on January 1, 2026. Part of the investment is held as inventory (finished trucks and raw parts valued at the bill-of-materials cost of \$0.57 per truck): Cash \$177,200 + Inventory \$22,800 = Total assets \$200,000 = Net worth. Selling price is \$2.50 per truck.

Transactions performed.

1. A sale of \$50,000 (20,000 trucks at \$2.50 to Cars are Us).
2. A purchase of \$20,000 of inventory (200,000 tanks at \$0.10 from Mr Mixers).
3. One payroll period for John Smith (\$150,000 per year, paid monthly).

The “after” statements of each transaction are the “before” statements of the next. An optional month-end close is included at the end to demonstrate the net-30 settlement.

Summary of changes

Stage	Cash	A/R	Inventory	A/P	Sales	Expenses	Net worth
Initial conditions	177,200.00	0.00	22,800.00	0.00	0.00	0.00	200,000.00
1. \$50,000 sale	177,200.00	50,000.00	11,400.00	0.00	50,000.00	0.00	238,600.00
2. \$20,000 purchase	177,200.00	50,000.00	31,400.00	20,000.00	50,000.00	0.00	238,600.00
3. Payroll	164,700.00	50,000.00	31,400.00	20,000.00	50,000.00	12,500.00	226,100.00
Month-end close	178,033.33	0.00	31,400.00	0.00	50,000.00	29,166.67	209,433.33

Key balances in dollars. Cost of goods sold is \$11,400.00 in every stage after the sale. The balance sheet balances at every stage (Assets = Liabilities + Net worth).

1 Initial conditions

Income statement and balance sheet

Print

Mr. Truck Inc., as of January 1, 2026

Income statement

Sales	
Sales	\$0.00
Cost of goods sold	\$0.00
Gross profit	\$0.00
Expenses	
Payroll	\$0.00
Payroll withholding	\$0.00
Bills (10 expense accounts)	\$0.00
Total expenses	\$0.00
Other income	\$0.00
Operating income	\$0.00
Income taxes	\$0.00
Net income	\$0.00

Balance sheet

Assets		Liabilities and net worth	
Cash	\$177,200.00	Accounts payable	\$0.00
Accounts receivable	\$0.00	Notes payable	\$0.00
Inventory	\$22,800.00	Accruals	\$0.00
Total current assets	\$200,000.00	Total current liabilities	\$0.00
Land/buildings	\$0.00	Mortgage	\$0.00
Equipment	\$0.00	Total long-term debt	\$0.00
Furniture and fixtures	\$0.00	Total liabilities	\$0.00
Total fixed assets	\$0.00	Owner investment	\$200,000.00
Total assets	\$200,000.00	Net income to date	\$0.00
		Net worth	\$200,000.00
		Total liabilities and net worth	\$200,000.00

Balanced: total assets equal total liabilities plus net worth.

Figure 1: Income statement and balance sheet at the initial conditions (January 1, 2026). No activity has been recorded, so every income statement line is zero. Total assets of \$200,000 equal net worth of \$200,000 (the owner's investment).

2 Transaction 1: Sale of \$50,000

An invoice was created for 20,000 trucks at \$2.50 each. Because terms are net 30, the sale is not yet cash.

Invoice history									Create invoice
Invoice #	Date	Due	Customer	Quantity	Price/unit	Total	COGS	Status	Paid on
1	2026-01-01	2026-01-31	Cars are Us	20,000	\$2.50	\$50,000.00	\$11,400.00	open	
Total sales		Total COGS							
\$50,000.00		\$11,400.00							

Figure 2: Invoice history: invoice #1 to Cars are Us for 20,000 units, \$50,000.00 total, due January 31, 2026, status open.

Income statement and balance sheet

Print

Mr. Truck Inc., as of January 1, 2026

Income statement

Sales	
Sales	\$50,000.00
Cost of goods sold	\$11,400.00
Gross profit	\$38,600.00
Expenses	
Payroll	\$0.00
Payroll withholding	\$0.00
Bills (10 expense accounts)	\$0.00
Total expenses	\$0.00
Other income	\$0.00
Operating income	\$38,600.00
Income taxes	\$0.00
Net income	\$38,600.00

Balance sheet

Assets		Liabilities and net worth	
Cash	\$177,200.00	Accounts payable	\$0.00
Accounts receivable	\$50,000.00	Notes payable	\$0.00
Inventory	\$11,400.00	Accruals	\$0.00
Total current assets	\$238,600.00	Total current liabilities	\$0.00
Land/buildings	\$0.00	Mortgage	\$0.00
Equipment	\$0.00	Total long-term debt	\$0.00
Furniture and fixtures	\$0.00	Total liabilities	\$0.00
Total fixed assets	\$0.00	Owner investment	\$200,000.00
Total assets	\$238,600.00	Net income to date	\$38,600.00
		Net worth	\$238,600.00
		Total liabilities and net worth	\$238,600.00

Balanced: total assets equal total liabilities plus net worth.

Figure 3: Statements after the sale. Before: Figure 1.

What changed:

- Sales +\$50,000.00 and Accounts receivable +\$50,000.00 (net 30; cash is unchanged).
- Cost of goods sold +\$11,400.00 (20,000 trucks × \$0.57) and inventory decreases from \$22,800.00 to \$11,400.00.
- Gross profit and net income are \$38,600.00, so net worth rises to \$238,600.00.

3 Transaction 2: Purchase of \$20,000 of inventory

A purchase order was created for 200,000 tanks at \$0.10 each. The parts are received immediately and billed on net-30 terms.

Purchase order history										Create PO
PO #	Date	Due	Supplier	Part	Ordered	Received	Price/part	Total	Status	Paid on
1	2026-01-01	2026-01-31	Mr Mixers	Tank	200,000	200,000	\$0.10	\$20,000.00	open	
Total purchased										
\$20,000.00										

Figure 4: Purchase order history: PO #1 to Mr Mixers for 200,000 tanks, \$20,000.00 total, due January 31, 2026, status open.

Income statement and balance sheet

Print

Mr. Truck Inc., as of January 1, 2026

Income statement

Sales	
Sales	\$50,000.00
Cost of goods sold	\$11,400.00
Gross profit	\$38,600.00
Expenses	
Payroll	\$0.00
Payroll withholding	\$0.00
Bills (10 expense accounts)	\$0.00
Total expenses	\$0.00
Other income	\$0.00
Operating income	\$38,600.00
Income taxes	\$0.00
Net income	\$38,600.00

Balance sheet

Assets		Liabilities and net worth	
Cash	\$177,200.00	Accounts payable	\$20,000.00
Accounts receivable	\$50,000.00	Notes payable	\$0.00
Inventory	\$31,400.00	Accruals	\$0.00
Total current assets	\$258,600.00	Total current liabilities	\$20,000.00
Land/buildings	\$0.00	Mortgage	\$0.00
Equipment	\$0.00	Total long-term debt	\$0.00
Furniture and fixtures	\$0.00	Total liabilities	\$20,000.00
Total fixed assets	\$0.00	Owner investment	\$200,000.00
Total assets	\$258,600.00	Net income to date	\$38,600.00
		Net worth	\$238,600.00
		Total liabilities and net worth	\$258,600.00

Balanced: total assets equal total liabilities plus net worth.

Figure 5: Statements after the purchase. Before: Figure 3.

What changed:

- Inventory +\$20,000.00 (from \$11,400.00 to \$31,400.00).
- Accounts payable +\$20,000.00, so total liabilities are \$20,000.00.
- The income statement does not change: buying inventory is not an expense. The cost reaches the income statement as COGS only when the trucks are sold.
- Total assets and total liabilities plus net worth both rise to \$258,600.00.

4 Transaction 3: One payroll period

John Smith (salary \$150,000 per year, single, one Illinois allowance) was paid one month of salary. Withholding uses 2026 rates: federal percentage method (IRS Pub. 15-T), Illinois 4.95%, Social Security 6.2% and Medicare 1.45%.

Pay employee

Pays one month of salary on January 1, 2026. Cash goes down by gross pay; net pay is recorded as Payroll and the taxes as Payroll Withholding.

Employee

John Smith

Bonus (\$, optional)

0

Pay employee

This month's paychecks

Employee	Gross	Federal	State (IL)	Social Security	Medicare	Net pay
John Smith	\$12,500.00	\$2,061.17	\$606.68	\$775.00	\$181.25	\$8,875.90

2026 rates: federal percentage method (IRS Pub. 15-T), Illinois 4.95% after IL-W-4 allowances, Social Security 6.2%, Medicare 1.45%. Bonuses are withheld federally at the 22% supplemental rate.

Figure 6: Pay employee form with the calculated paycheck: gross \$12,500.00, federal \$2,061.17, Illinois \$606.68, Social Security \$775.00, Medicare \$181.25, net pay \$8,875.90.

Paid John Smith: cash -\$12,500.00; Payroll +\$8,875.90; Payroll Withholding +\$3,624.10.

Payroll payments and history

Pay employee

Date paid	Employee	Disbursement	Withholding
2026-01-01	John Smith	\$8,875.90	\$3,624.10

Total disbursement	Total withholding	Total gross payroll
\$8,875.90	\$3,624.10	\$12,500.00

Tax detail

Date	Salary	Bonus	Federal tax withheld	State tax withheld	Social Security	Medicare	Amount paid
2026-01-01	\$12,500.00	\$0.00	\$2,061.17	\$606.68	\$775.00	\$181.25	\$8,875.90

Figure 7: Payroll history recording the payment, with total disbursement \$8,875.90, total withholding \$3,624.10 and the tax detail.

Income statement and balance sheet

Print

Mr. Truck Inc., as of January 1, 2026

Income statement

Sales	
Sales	\$50,000.00
Cost of goods sold	\$11,400.00
Gross profit	\$38,600.00
Expenses	
Payroll	\$8,875.90
Payroll withholding	\$3,624.10
Bills (10 expense accounts)	\$0.00
Total expenses	\$12,500.00
Other income	\$0.00
Operating income	\$26,100.00
Income taxes	\$0.00
Net income	\$26,100.00

Balance sheet

Assets		Liabilities and net worth	
Cash	\$164,700.00	Accounts payable	\$20,000.00
Accounts receivable	\$50,000.00	Notes payable	\$0.00
Inventory	\$31,400.00	Accruals	\$0.00
Total current assets	\$246,100.00	Total current liabilities	\$20,000.00
Land/buildings	\$0.00	Mortgage	\$0.00
Equipment	\$0.00	Total long-term debt	\$0.00
Furniture and fixtures	\$0.00	Total liabilities	\$20,000.00
Total fixed assets	\$0.00	Owner investment	\$200,000.00
Total assets	\$246,100.00	Net income to date	\$26,100.00
		Net worth	\$226,100.00
		Total liabilities and net worth	\$246,100.00

Balanced: total assets equal total liabilities plus net worth.

Figure 8: Statements after the payroll. Before: Figure 5.

What changed:

- Cash $-\$12,500.00$ (gross pay), from $\$177,200.00$ to $\$164,700.00$.
- Payroll expense $+\$8,875.90$ (net pay) and Payroll withholding $+\$3,624.10$ (taxes), for total expenses of $\$12,500.00$.
- Net income falls to $\$26,100.00$ and net worth to $\$226,100.00$. Accounts receivable and payable are unchanged.

5 Optional: Month-end close (net-30 settlement)

At the end of the month the books are updated by posting receivables to cash and paying the bills, even if 30 days have not quite passed. The ten fixed monthly expense accounts (\$200,000 per year) are also paid.

Testing and setup

Close the month

Current date: **2026-01-01**. Closing will:

Collect 1 open invoice(s)	\$50,000.00
Pay 1 open purchase order(s)	\$20,000.00
Pay the 10 monthly bills (if checked)	\$16,666.67

☒ Pay monthly bills

Then the date advances to **2026-02-01**.

Close month

Figure 9: Month-end close: collect the \$50,000.00 invoice, pay the \$20,000.00 purchase order and pay \$16,666.67 of monthly bills, then advance the date to February 1, 2026.

Income statement	
Sales	
Sales	\$50,000.00
Cost of goods sold	\$11,400.00
Gross profit	\$38,600.00
Expenses	
Payroll	\$8,875.90
Payroll withholding	\$3,624.10
Bills (10 expense accounts)	\$16,666.67
Cleaning	\$833.33
Donuts	\$833.33
Electricity	\$2,500.00
Gas	\$2,500.00
High Speed Internet	\$1,666.67
Maintenance	\$2,500.00
Phone	\$1,666.67
Sewage	\$1,666.67
Travel	\$1,666.67
Water	\$833.33
Total expenses	\$29,166.67
Other income	\$0.00
Operating income	\$9,433.33
Income taxes	\$0.00
Net income	\$9,433.33

Balance sheet	
Assets	
Cash	\$178,033.33
Accounts receivable	\$0.00
Inventory	\$31,400.00
Total current assets	\$209,433.33
Land/buildings	\$0.00
Equipment	\$0.00
Furniture and fixtures	\$0.00
Total fixed assets	\$0.00
Total assets	\$209,433.33
Liabilities and net worth	
Accounts payable	\$0.00
Notes payable	\$0.00
Accruals	\$0.00
Total current liabilities	\$0.00
Mortgage	\$0.00
Total long-term debt	\$0.00
Total liabilities	\$0.00
Owner investment	\$200,000.00
Net income to date	\$9,433.33
Net worth	\$209,433.33
Total liabilities and net worth	\$209,433.33
Balanced: total assets equal total liabilities plus net worth.	

Figure 10: Income statement (left) and balance sheet (right) after the month-end close, as of February 1, 2026. Before: Figure 8.

What changed:

- Accounts receivable \$50,000.00 → \$0 and accounts payable \$20,000.00 → \$0.
- Cash: $\$164,700.00 + \$50,000.00 - \$20,000.00 - \$16,666.67 = \$178,033.33$.
- Bills of \$16,666.67 are added to expenses (total \$29,166.67), leaving net income of \$9,433.33 and net worth of \$209,433.33.

Program code

The complete source code (app.py, accounting.py, payroll.py, db.py, templates and stylesheet) is submitted separately as `mr_truck_books.zip`.